Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Brady, Joseph V. Ph.D.		
Project Title:	Psychosocial Performance Factors in Space Dwelling Groups		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI Dr. Joseph Brady passed away in July 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert (The Johns Hopkins University School of Medicine) Hursh, Steven (Institutes for Behavior Resources, Inc.)		
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	Despite uncertainties regarding the performance requirements of projected space laboratories, work stations, interplanetary vehicles and settlements beyond the Earth’s atmosphere, a common feature of spaceflight endeavors over the next half century will be extended stays by human groups in extraterrestrial habitats. The imperatives and opportunities associated with configuring effective psychosocial performance models in support of extended stays by human groups in extraterrestrial habitats are best served by research approaches that are both heuristic and innovative. The development of functional human and ecological models for such space dwelling groups must, in the first instance, be based upon sound scientific principles with research objectives focused upon the management of semi-permanent as well as permanent groups involved in both operational and space science missions. Research methodologies provided by distributed interactive multi-person simulations in computer generated environments are experimental test beds for		

Task Description:	modeling psychosocial performances within and between space-dwelling and Earth- based groups. They provide automated means of setting the context for analysis of performance in space-dwelling groups and monitoring electronically the effects of varying communication and other experimental conditions. Communication constraints impair the ability to accurately discern and interact with the realities of mission operations. Distributed interactive simulation experiments can characterize the effects of variations in the structure and function of communication channels within and between space-dwelling and Earth-based groups as well as the effects of stressful environmental and behavioral interactions upon psychosocial performance effectiveness. Simulation experiments also determine the effects of variations in the appetitive/aversive characteristics of incentive systems as well as the effects of selection, training, and experience within and between space-dwelling and Earth-based groups. Conceptual and methodological advances that effectively promote psychosocial and ecological stability will ultimately benefit larger societal units, including those that remain Earth-bound, by enhancing an educational and training technology based upon an expanded generalizeable knowledge base. The results of these studies with scenario tasks requiring identification of geologic samples designated by different rules showed clearly that cooperative and productive psychosocial interactions could be maintained between individually isolated and dispersed crewmembers in the simulated task environment. All experimental flight crews actively engaged in communicating and effective problem-solving over extended time intervals without benefit of one another's physical presence. In addition, the investigations of communication modality constraints indicated that there was a high degree of interchangeability between the available communication modes. This functional interchangeability between available modes of communication was shown to be an effective countermeasure to communication modality constraints as was the use of 'indexical' features of the simulation display to maintain effective crew performance. These studies have been extended to the analysis of crew configuration changes in simulated space-dwelling groups. A modified simulation scenario employed the standard Orbiter vehicle and two Lander/Rover hybrid vehicles, each of which possessed the combined capabilities of the standard Lander and Rover vehicles. Variations in crew configuration consisted of crewmate reassignments to simulated vehicles other than those normally operated. These reassignments were based upon baseline performance levels and resulted in variations in crewmate skill levels for operating the different vehicles. Operation of the Orbiter for example, required successful solving of the rules for efficient collection of grade valued geologic specimens making it possible to differentiate changes that resulted in "Strong" and "Weak" crew configurations. An embedded measure of the Orbiter efficiency was developed based upon the "lag time" between the receipt of geologic specimen collection data by the Orbiter and the response submitting the specimen for analysis. Analysis "lag times" were tallied in 3 second time bins to form an Orbiter analysis lag time histogram. The distribution of lag times indicated the extent to which the Orbiter processed specimen data in a timely manner; the greater the proportion of specimens processed with short lag times, the higher the efficiency of data throughput. Replications of configuration change experiments with several simulated spaceflight crews indicated clearly that baseline performance effectiveness levels under experienced "Strong" Orbiter conditions were significantly reduced during missions following crew assignment changes that resulted in "Weak" Orbiter configurations. Although several other performance measures (e.g. specimen collection rates, audio, text, and whiteboard messaging rates) showed no consistent effects as a function of such configuration changes, analysis of "lag time" data provided an effective account of the observed performance changes. With experienced crews, lag times seldom exceeded 30 seconds under "Strong" Orbiter configurations with few lags over 1 minute and no lags exceeding 2 minutes. Under "Weak" Orbiter configurations by contrast, lag times frequently exceeded 2 minutes with several lags greater than 3 minutes. The experiments to be conducted during the coming year will extend the distributed interactive simulation studies to analyze the effects of selective communication between crew members and mission control as these are influenced by changes in crew configuration and incentive conditions. The analyses will focus upon both individual crew member performance and psychosocial changes in the patterns of distributed interactive communication.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research conducted within the context of distributed interactive simulation models will provide the basis for developing effective patterns of communication and problem solving strategies as well as a range training procedures to enhance problem solving effectiveness. The Earth-based applications of this research will extend to the small operational group selection and training process as well as the management of stressful interactions and the maintenance of group cohesion and productivity. Not only can the outcome of these studies be expected to have an important impact on safety and the quality of life in many Earth-based applied settings, but larger societal units will ultimately benefit from the resulting conceptual and methodological advances that effectively promote social and ecological stability while concurrently enhancing an education and training technology that assures effective communication of an expanded generalizeable knowledge base.
Task Progress:	Task progress during the reporting year has focused on the experimental analysis of crew configuration changes in simulated space-dwelling groups. A modified simulation scenario employed the standard Orbiter vehicle and two Lander/Rover hybrid vehicles, each of which possessed the combined capabilities of the standard Lander and Rover vehicles. Variations in crew configuration consisted of vehicle reassignments to simulated vehicles other than those normally operated. Reassignments were based upon baseline performance levels and resulted in variations in crewmate skill levels for operating the different vehicles. Operation of the Orbiter for example, required successful solving of the rules for efficient collection of grade valued geologic specimens with the indicated changes resulting in "Strong" and "Weak" crew configurations. An embedded measure of the Orbiter efficiency was developed based upon the "lag time" between receipt of the geologic specimen collection data by the Orbiter and the response submitting the specimen for analysis. Analysis "lag times" were tallied in 3 second time bins to form an Orbiter analysis lag time histogram. The distribution of lag times indicated the extent to which the Orbiter processed the specimen data in a timely manner; the greater the proportion of specimens processed with short lag times the higher the efficiency of data throughput. Data collected from three separate crews (AR, BE, and BF) thus far completing several replications of crew configuration change experiments indicate clearly that baseline performance effectiveness levels under experienced "Strong" Orbiter conditions were significantly reduced during simulated spaceflight missions following crew assignment changes that resulted in "Weak" Orbiter configurations. Although several other performance measures (e.g. specimen collection rates, audio, text, and whiteboard messaging rates) showed no consistent effects as a function of such configuration changes, analysis of the "lag time" data provided an effective account of the observed performance decrements. With Crew AR for example, lag times seldom exceeded 30 seconds under "Strong" Orbiter configurations with few lags over 1 minute and no lags exceeding 2 minutes. Under "Weak" Orbiter configurations by contrast, lag

	times frequently exceeded 2 minutes with several lags greater than 3 minutes. Similarly with Crew BE, “Strong” Orbiter configuration lag times seldom exceeded 60 seconds with only rare occurrences of 2 to 3 minute delays. With Crew BE “Weak” Orbiter configurations, initial lag times were predominantly longer that 3 minutes although some reduction in lag time delays was observed with repeated simulated spaceflight missions under “Weak” Orbiter conditions. The effects observed with Crew BF further confirmed the value of the embedded “lag time” measure in assessing crew configuration performance levels under “Strong” and “Weak” Orbiter conditions.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
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